

Azeem Husain

azeem.husain@hotmail.com | (615) 838-2817 | Willing to Relocate

Skills

CAD: SolidWorks, CATIA V5, CATIA V6 (3DEXPERIENCE), Onshape, Fusion 360, AutoCAD, Blender

Analysis: MATLAB, SolidWorks FEA, Ansys Speos

Prototyping: 3D Printing, Metal fabrication, Circuits, Arduino programming

Experience

Mechanical Design Engineer, Honda – Raymond, OH

Aug 2023 – Present

- Developed 3D CAD layouts and issued both concept and production drawings for exterior lighting assemblies across five Honda and Acura projects using CATIA V5 and CATIA V6 (3DEXPERIENCE).
- Designed injection-molded plastic parts using DFM/DFA principles and GD&T based internal tolerancing standards to ensure manufacturability, dimensional robustness, and repeatable high-volume production.
- Increased concept approval rate by 50% by translating styling studio intent and strict regulatory constraints into manufacturable, cost-effective design proposals.
- Doubled my departments supplier base through new supplier sourcing and cost analysis.
- Boosted departmental task efficiency by as much as 200% by developing internal automation tools with Power Automate and Excel VBA to streamline repetitive workflows.
- Conducted market research and competitor benchmarking to identify design improvement opportunities, inform design targets, and improve future product competitiveness.

Fluid Dynamics Lab Research Assistant, University of Tennessee – Knoxville, TN

Aug 2022 – Dec 2022

- Designed precision test fixtures for fluid dynamics experiments using SolidWorks.
- Conducted data acquisition, analysis, and validation using MATLAB and Excel.
- Co-authored a research paper published in *Physics of Fluids*.

Process Engineering Co-op, DENSO – Maryville, TN

May 2022 – Aug 2022

- Designed production equipment countermeasures in SolidWorks and AutoCAD, reducing production line downtime by 15% and unplanned stoppages by 30%.
- Facilitated the launch of a fully automated production line for automotive electrical hardware.
- Worked closely with production equipment manufacturers both domestically and globally to ensure adherence to design specifications and on-time delivery.

Projects

BattleBots and Robot Battles

- Designed, fabricated, and iteratively implemented countermeasures on competitive combat robots including structural design, drivetrain design, material selection, and failure analysis.
- Ranked: #3 Fairyweight, top 30 Beetleweight, and top 50 Antweight nationally.
- Prepared and presented educational demonstrations for the Boy Scouts of America, Nashville Maker Faire, Nashville's Adventure Science Center, and the University of Tennessee.
- Volunteered as a judge and event coordinator for multiple competitions including the National Robotics Challenge.

Baja SAE

- Performed chassis and drivetrain design in SolidWorks, applying FEA to optimize weight, strength, and performance.
- Utilized 3D printing, CNC machining, and welding to manufacture a competition-ready vehicle.

Education and Certifications

University of Tennessee Knoxville – BS in Mechanical Engineering

May 2023

- Graduated Magna Cum Laude (3.66 GPA)
- Vice President of ASME Student Chapter

Certified SolidWorks - Associate in Mechanical Design (CSWA)

Nov 2022